

```
#include <stdio.h>
```

```
int book_status[5] = {1, 1, 0, 1, 0};
```

```
char book_titles[5][30] = {"C Programming", "Algorithms", "Data Structures", "Web Dev", "Physics"};
```

```
void displayBooks() {
```

```
    int i;
```

```
    printf("\n--- Available Books ---\n");
```

```
    for (i = 0; i < 5; i++) {
```

```
        printf("%d. %s [%s]\n", i + 1, book_titles[i], book_status[i] ? "AVAILABLE" : "ISSUED");
```

```
    }
```

```
}
```

```
int getTotalBooks() {
```

```
    return 5;
```

```
}
```

```
void borrowBook(int bookIndex) {
```

```
    if (bookIndex >= 1 && bookIndex <= 5) {
```

```
        if (book_status[bookIndex - 1] == 1) {
```

```
            book_status[bookIndex - 1] = 0;
```

```
            printf("\nBook '%s' has been successfully borrowed.\n", book_titles[bookIndex - 1]);
```

```
        } else {
```

```
            printf("\nBook '%s' is already issued.\n", book_titles[bookIndex - 1]);
```

```
        }
```

```
    } else {
```

```
        printf("\nInvalid book selection.\n");
```

```
    }
```

```
}
```

```
float calculateFine(int daysLate) {  
    float fine_rate = 5.0;  
    if (daysLate < 0) daysLate = 0;  
    return (float)daysLate * fine_rate;  
}
```

```
int main() {  
    int days, book_choice;  
    printf("\nName :- BHARGAV GUJJAR \n");  
    printf("\n--- Library Management System ---\n");  
  
    displayBooks();  
  
    printf("\nTotal number of books in inventory: %d\n", getTotalBooks());  
  
    printf("\nEnter book number to borrow: ");  
    scanf("%d", &book_choice);  
    borrowBook(book_choice);  
  
    printf("Enter number of days overdue: ");  
    scanf("%d", &days);  
    printf("Fine for %d days: Rs. %.2f\n", days, calculateFine(days));  
  
    displayBooks();  
  
    return 0;  
}
```